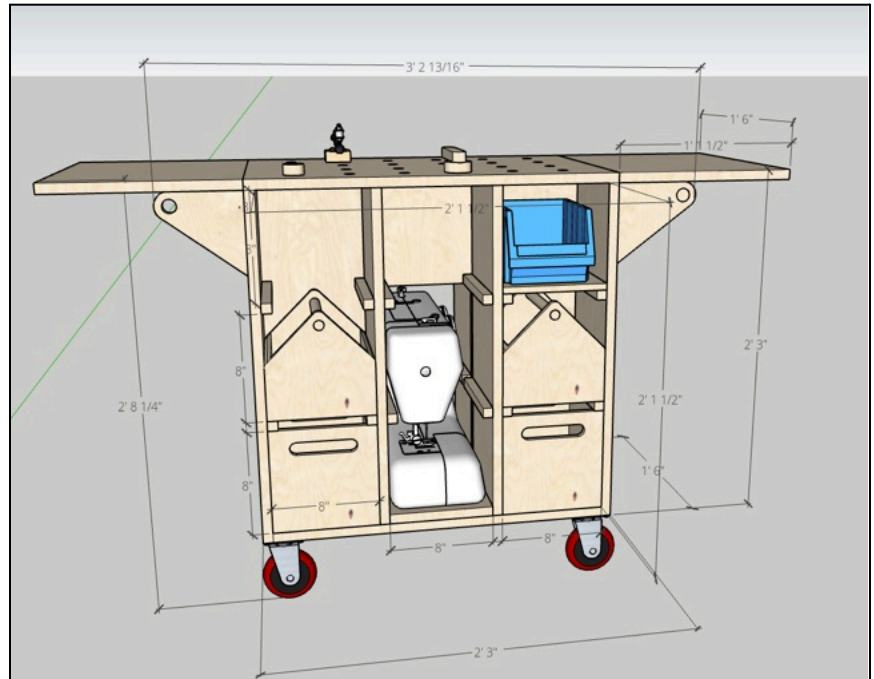


Fix It Cart Build Plan

Build a Fix It Cart for elementary and middle school students to use in support of their repair club or other repair activities around school. There are two variations of this plan, one for larger students (pictured here) and another shorter version which is suitable for younger students.

This design is easily customizable and, being made of wood, can have hooks and add-ons screwed into the front and back to hang tools, cords, etc. as needed. The drawers are boxes, but can easily be substituted by simple boards to make shelves on which anything can be stored.



Materials List

Note: A list of example parts will be included below, but any equivalent materials can be used as needed)

Material	Qty	Material	Qty
Sheet of $\frac{3}{4}$ " birch plywood (4' x 8')	1	5" lockable casters (or 4" will work) with mounting screws. At least 2 of them must swivel to enable steering.	4
1" hardwood (e.g., maple or oak) dowel, 18" length	1	1 $\frac{1}{4}$ " Pocket screws (Kreg or equivalent)	32 min.
1 $\frac{1}{4}$ " finish nails	48 min.	180 degree Brass Hinges: Sewing machine table hinges also known as Butler tray hinge with round edge. 2.8" L x 1.2" W x 0.1" T or equivalent	4
1 $\frac{1}{2}$ " Wood #10 wood screws	4	1 $\frac{1}{2}$ " #8 wood screws	10
Sacrificial boards to create a cutting surface across the sawhorses. E.g, some 2" x 4" x 8' are great for this purpose.	4 min.	3" x 2" Brass utility hinge with screws	2
Adhesive velcro squares or circles (used to hold the Eave supports against the End when not in use)		1 pint can of wipe-on polyurethane	1

Materials for the Drawer Boxes

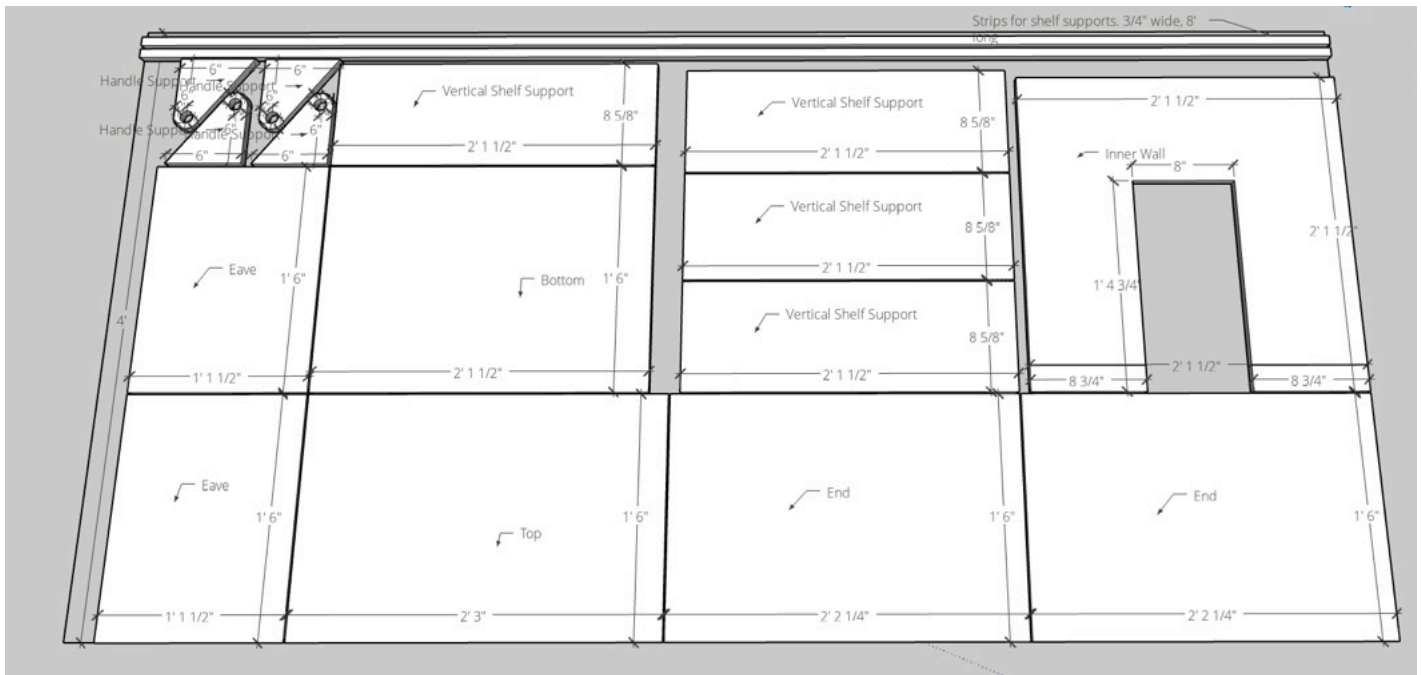
Material	Qty	Material	Qty
Sheet of ½" birch plywood (4' x 8')	1	1" #6 screws	16 per box (288 if making all 18 boxes)

Tools Needed

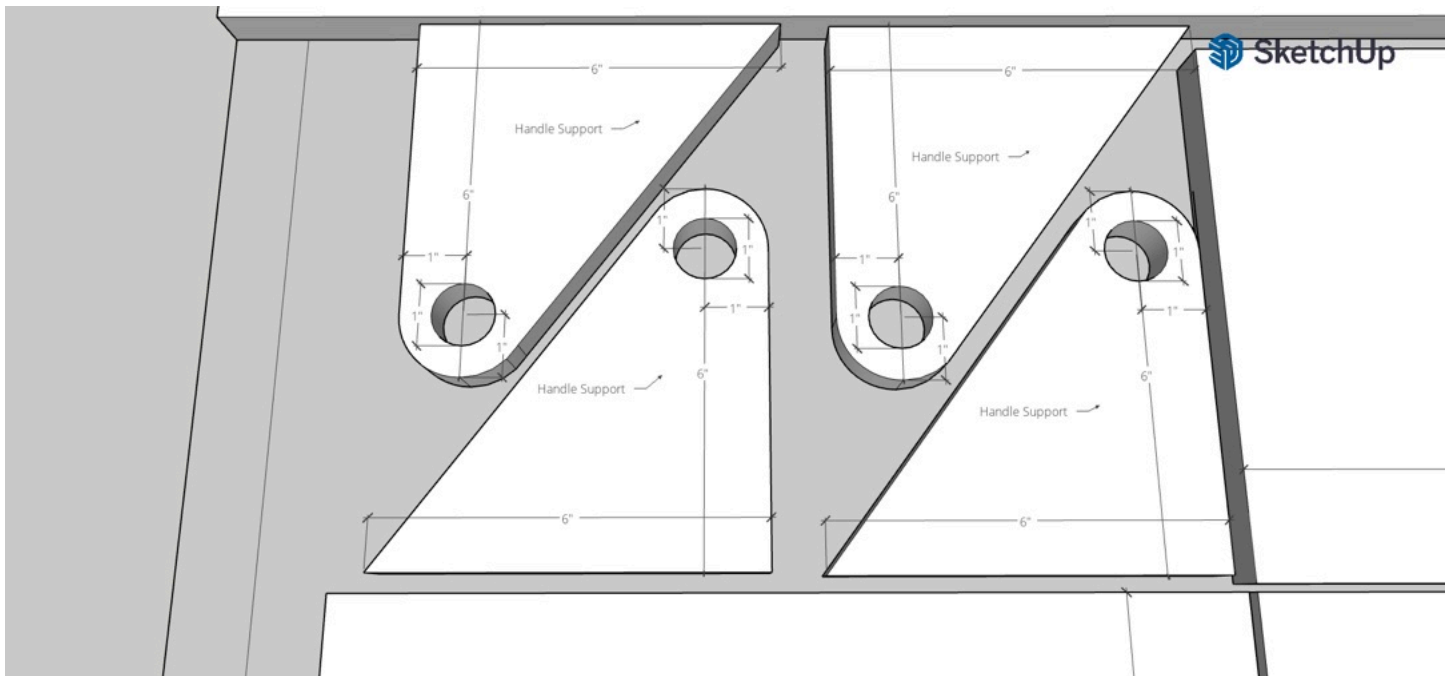
Circular saw with plywood blade (fine toothed)	Handsaw or jig saw for cutting interior corners
Miter saw (optional, for cutting support pieces)	Drill and standard drill bit set including a ⅛" drill bit
Pocket screw kit	Screwdriver or Power driver with appropriate bits for pocket screws and philips heads. You will likely need a short/stubby Phillips to screw on the handle supports.
¾" forstner bits (optional, for dog holes on top of cart)	1" forstner bit (for handle dowel holes)
Router or wood chisel (for cutting the eave hinge recesses)	Suitable workstation or saw horses
Cutting straight edge 8' long (can use factory edge of plywood if needed)	2 C-Clamps for holding straight edge during cutting
Hammer	Saw horses (2)
Belt sander (optional, but helpful for sanding the curve on the handle/eave supports)	Carpenter's square
Sandpaper, recommend 1 sheet of 180 grit and a sheet of 400 grit for finishing work.	

Cut Sheet

You can cut all the pieces of the cart out of a single sheet of 8' x 4' plywood using the following cut sheet as the layout. **Note that you need to make each cut separately and then measure and cut the next piece. If you don't do this, the kerf of the saw blade will throw off your measurements.**



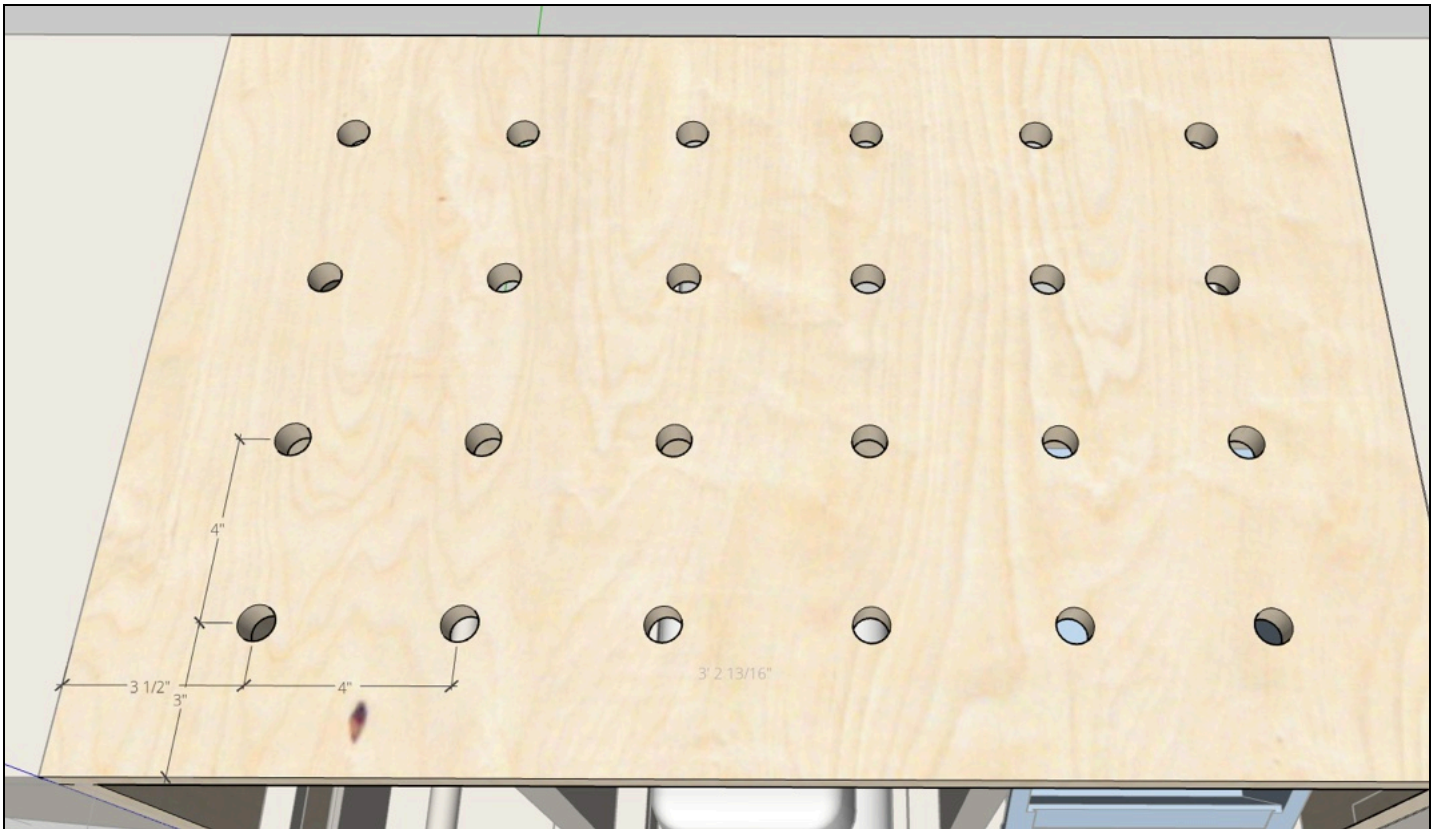
Detail for the eave/handle supports



Build Instructions

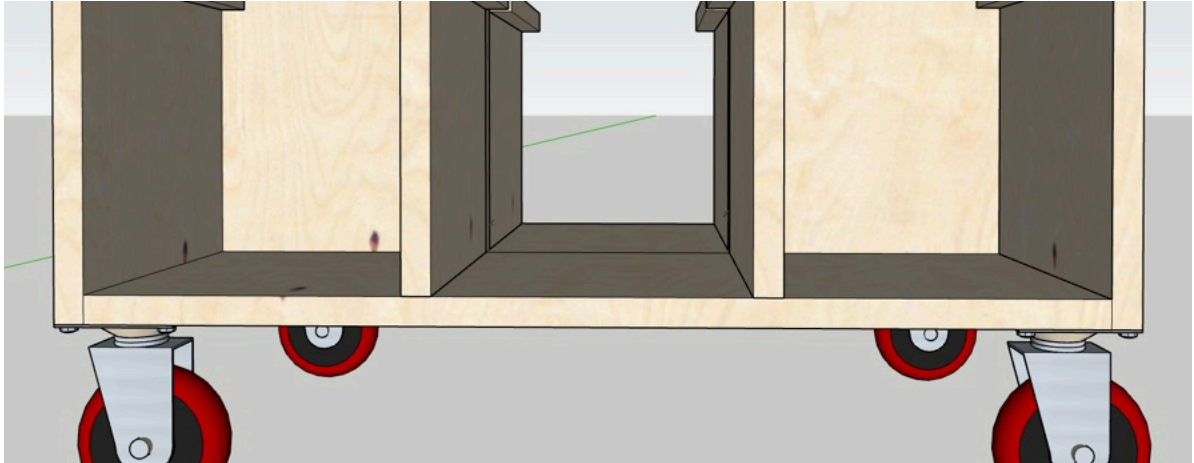
1. Following the cut sheet above, cut all of the pieces out of the sheet of $\frac{3}{4}$ \" birch plywood

- a. Be sure to have the sheet securely placed on top of some sacrificial 2" x 4" boards which are laid between the saw horses. As you work, be sure these don't move and check them often to ensure no hazard is created. If it's possible to secure the 2" x 4" boards to the saw horses, that is highly recommended.
 - b. Don't layout the entire sheet all at once. Proceed one cut at a time and after each cut is made, then measure for the next cut. This will avoid a common mistake of failing to account for the thickness of the sawblade (called the kerf).
 - c. Using the straight edge, c-clamps and a circular saw, cut out the pieces to the provided dimensions. Be precise. Cut to the line such that the kerf of the saw blade is to the exterior of the part you are cutting. Failure to do that will skew the dimensions of your finished pieces. Use a handsaw or jig saw to cut the interior corners of the cutout of the Inner Wall.
 - d. Note, when cutting the handle/eave supports (the triangular pieces), just cut them straight at first and then add the curve afterward with a jig saw or file or belt sander after you've drilled the holes in the next step.
 - e. Tip: use masking tape to label each piece using the same labels from the cut sheet above as you cut each piece out. This will prevent confusion about which piece is which later on.
2. Measure and drill out the holes for the handle/eave supports.
 - a. Reference the Detail for eave/handle supports in the Cut Sheet section above.
 - b. The center of the circles is the intersection of two lines 1" from each edge.
 - c. Be sure to drill straight perpendicular to the surface. Use a carpenter's square or other known square as a reference while drilling. Forstner bits are also wide enough that with care, you can tell if you are perpendicular during the drill.
 - d. Using the 1" forstner bit, drill holes in at least 2 of the supports. You only need a handle on one side of the cart and thus only 2 of the supports need holes to receive the dowel handle.
 - e. Once the holes are drilled, using a jig saw or belt sander, round off the corner so as not to have a sharp angle.
 3. Layout and drill the dog holes for the Top (optional). These holes enable adding dogs and clamps to hold workpieces on the Top. This is entirely optional and can be omitted if desired and you prefer to have a smooth Top. Be sure to follow the directions on your pocket screw kit carefully and set the drill depth correctly for $\frac{3}{4}$ " board thickness.
 - a. Layout a 4-by-6 grid, with 4" between each hole center in both axes. (see picture below).
 - b. Using the $\frac{3}{4}$ " forstner bit, drill out each hole (24 holes in all). Be sure to drill from the top surface (your plywood may have one surface which is nicer than the other) to the bottom so any tearout doesn't impact the appearance and smoothness of the top surface.



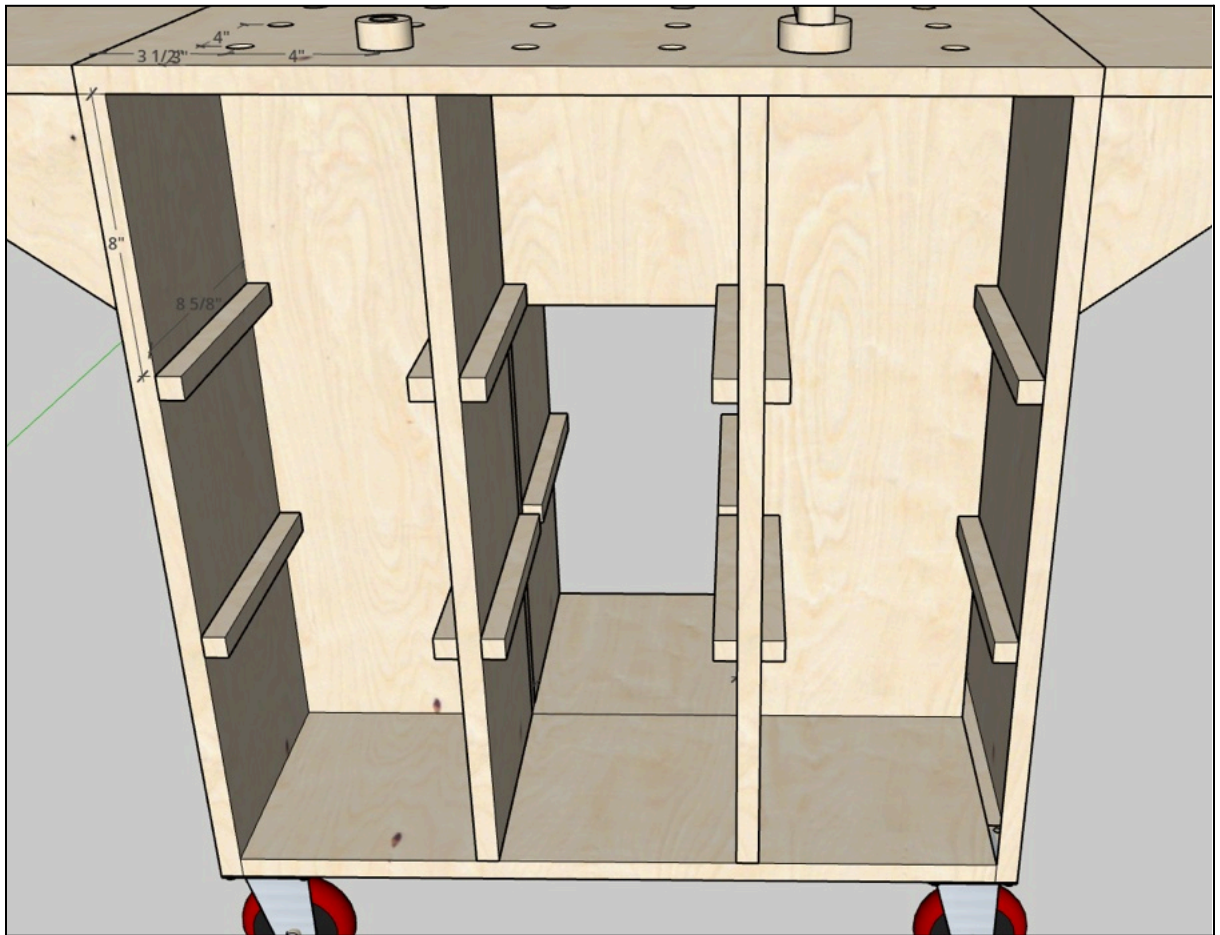
4. Cut the drawer supports to length.
 - a. From the cut sheet, you should have 2 long $\frac{3}{4}$ " wide strips. Using a miter saw or hand saw, cut those into 8 $\frac{1}{2}$ " lengths. If they are a little short, it isn't a problem. You should end up with 24 holes in total.
5. Pre-drill the pocket screw holes
 - a. First, take a moment to note which sides of your pieces are the tops and bottoms and which are the sides. The cart is overall 18" wide and 2' 3" (27") tall. Thus, it is taller than it is wide. That will tell you which edges to drill for screws to hold the sides to the Top and where to drill the Bottom to attach to the sides.
 - b. Using the pocket screw jig from the kit, drill holes as follows for each type of piece
 - i. Top - do not drill any pocket holes. It will receive screws from the sides and Vertical Supports.
 - ii. Ends - these are the front and back of the cart. Drill 4 equally spaced holes along one of the 18" edges of each. These holes will be used to attach the sides to the Top.
 - iii. Bottom - Drill 4 equally spaced holes along both of the 18" sides. These will be used to screw the Bottom to the 2 sides.
 - iv. Inner Wall - Drill 4 equally spaced holes along the 3 full length sides. These will be used to hold it to the Top and to the two Ends. Then, drill 2 equally spaced holes along both of the shorter bottom edge (on either side of the cutout). These will be used to screw the Inner Wall to the Bottom.
 - v. Vertical Supports (all 4 of them) - Drill 2 equally spaced holes along both short edges. These will be used to screw into the Top and Bottom of the cart. There is no need for screws to attach them to the center wall.
6. Assemble the wooden pieces to form the cart

- a. Using the proper driver for pocket screws and the 1 ¼" pocket screws, attach the pieces using the holes you've pre-drilled. It is best to have the pieces securely braced (e.g., against a wall) in the correct orientation to each other before drilling to avoid movement while screwing them together. Use assistants, clamps or whatever you have to accomplish this.
- b. No screws should be visible on the Top or on the front and back of the cart. All screw holes should be on the interior of the cart.
- c. Attach the Bottom to the two Ends (front and back). Note that the Bottom piece goes "between" the two sides, not "beneath" them as shown in the image below.



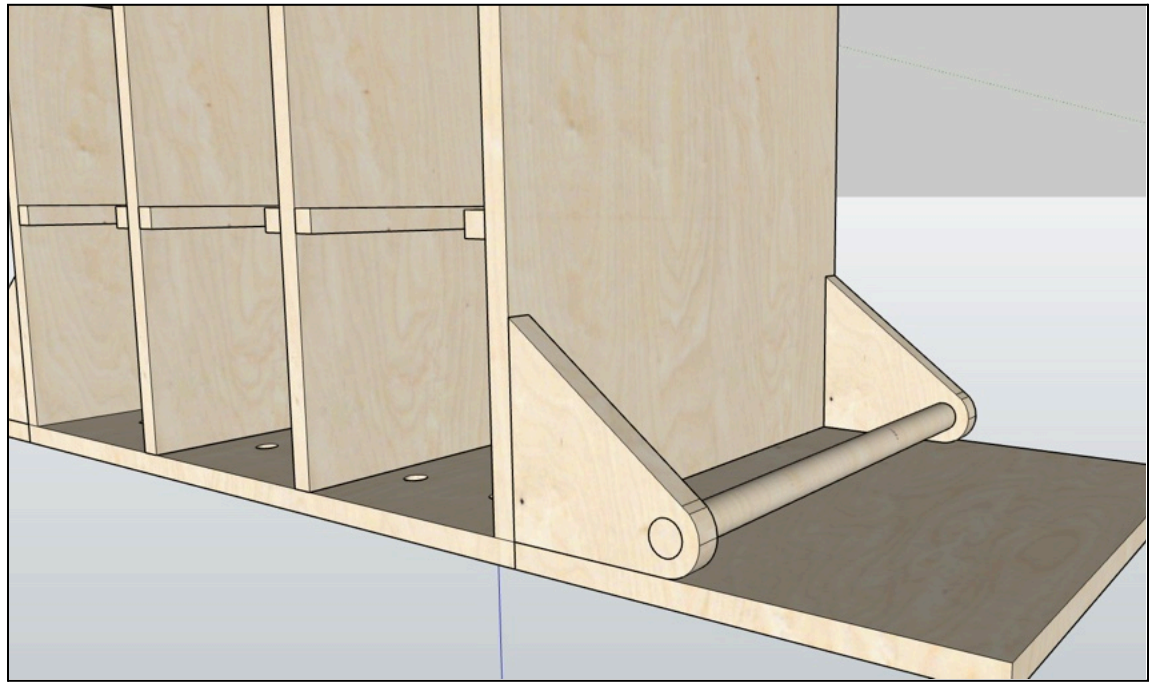
- d. Attach the Top to the two sides by screwing the sides into the underside of the Top through the pre-drilled holes.
- e. Attach the Shelf Supports to the Vertical Supports using a hammer and the finish nails. It's easier to do this now than to attempt it later when there isn't enough room to swing the hammer.
 - i. Attach to the Vertical Supports: Measure up 8 inches from the bottom and use a carpenter square to mark a perpendicular line. Align the bottom edge of the Shelf Support to that first line and nail it in. Repeat that for the other side. Then, mark another line 8" from the top of the first shelf support and align the bottom of the next shelf support to that and nail it in. You should end up with 3 equal spaces of 8" between each support on both sides of all 4 Vertical Supports (see the images to get the right idea).
- f. Attach the first two Vertical Supports on one side of the cart. They should each be 8" from the front or back respectively. They should be flush with the outer edge of the Top and Bottom boards. Screw them into the Top and Bottom using the predrilled holes.
- g. Lay the box with the 2 installed Vertical Supports on the floor or working surface. Insert the Inner Wall into the box, laying it on top of the Vertical Supports. Screw it into the Top, Bottom and two Ends using the pre-drilled holes.
- h. Attach the remaining two Vertical Supports identically to the first two so you end up with a symmetrical installation on both sides of the Inner Wall.
- i. Nail the remaining Shelf Supports to the two Ends following the same procedure you did previously for the Vertical Supports - measure up 8" from the base of each wall, use a square and then nail in the support. Nail in another 8" above the top of that Shelf Support. The result

should be an even set of shelf supports as shown in the pictures.

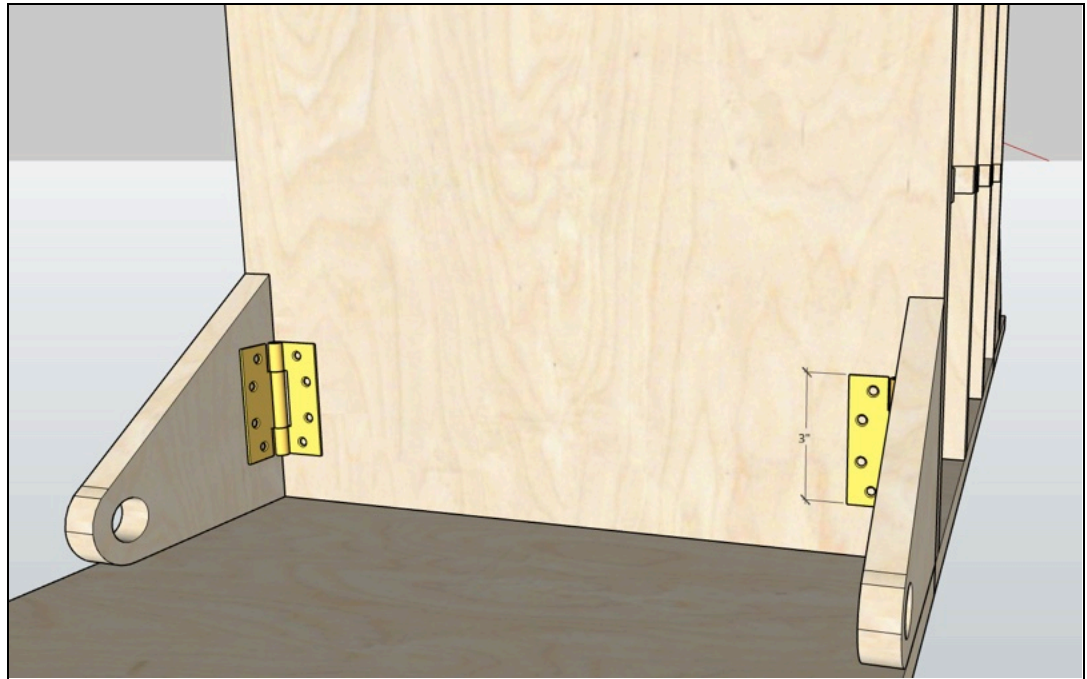


- j. Attach the handle supports
 - i. Note: There are 4 identical pieces which are labelled Eave/Handle supports. These serve a dual purpose of being the Handle and Eave Support on one side of the cart (the one users will push from). On that side, a dowel will run through the holes to form the handle. On the other side, the Eave Supports will be attached with hinges so they can fold in and out when needed to support the Eave in its unfolded position.
 - ii. To attach the Handle Supports, first, position the cart on the Top so that the Bottom is facing upward. Next, put one of the Eaves on the floor/work surface next to the Top where it will be once installed. This enables you to very precisely position the handle

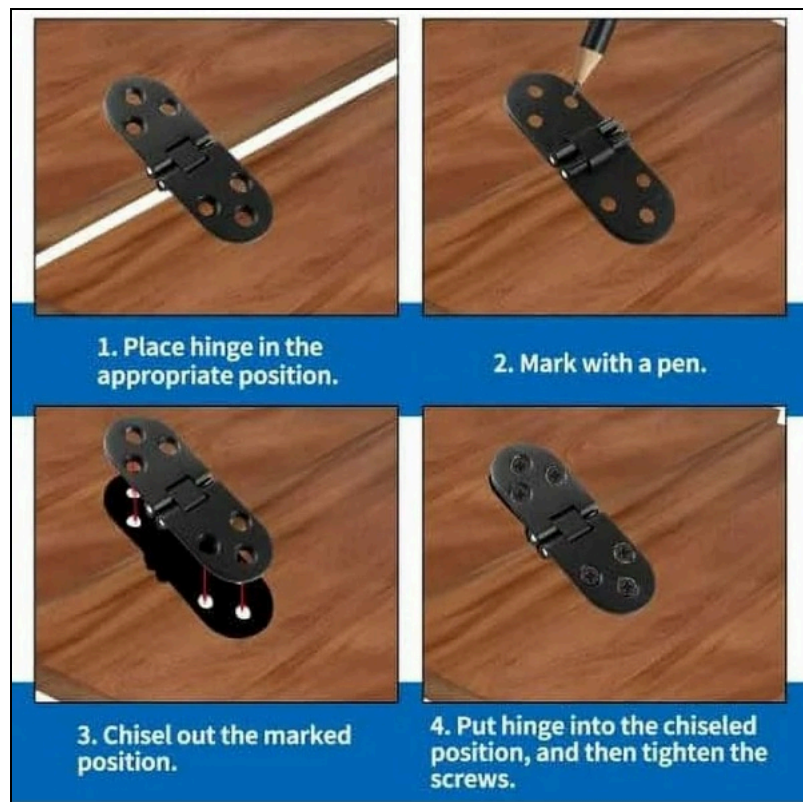
support just against the eave as shown.

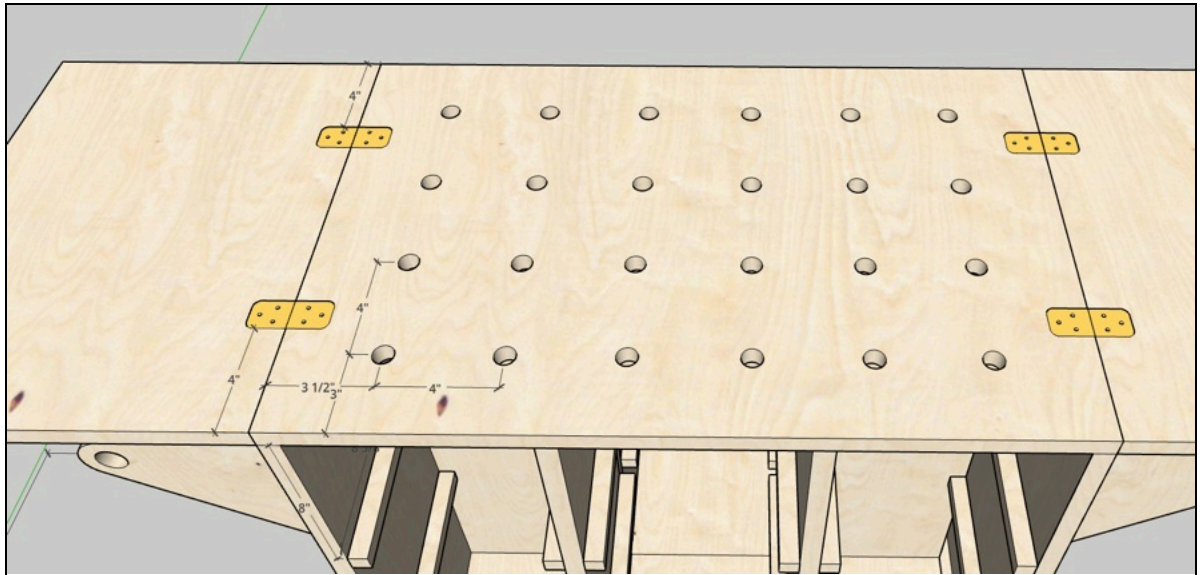


- iii. Mark the position of the Handle Supports with a pencil and then set them aside. Pre-drill two $\frac{1}{8}$ " holes where the center of the two supports will be from the outside of the wall toward the inside. The holes should be $1\frac{1}{2}$ " from the bottom and top of where the Handle Support meets the End wall. Then, reposition the Handle Supports and drill from the interior through the holes into the Handle Supports. This should create the four predrilled holes into which the #10 screws will be screwed to attach the handles firmly.
- iv. While the cart is still upside down, insert the 1" x 18" wooden dowel into the holes. It may be a tight fit. If so, use the 180 grit sandpaper to reduce the size of the towel where it will enter the holes until it will fit on both sides. Put the dowel into place in both holes to ensure alignment, centered nicely, and then use a Phillips screwdriver to screw the handle supports into place.
- v. To secure the down into the handles, predril a $\frac{1}{8}$ " hole from the bottom side of the support into the down on both supports. Then, screw in the #8 $1\frac{1}{2}$ " screws, one per side, into the dowels to hold them in place.
- vi. While the cart is still upside down, lay the other Eave next to the top on the other side of the cart and attach both supports in a similar position as to the handle size, however in this case you will attach them via the utility hinges as shown. Use the screws provided by the hinge manufacturer to attach them. No predrilling should be necessary, but it is good to do so if you can.

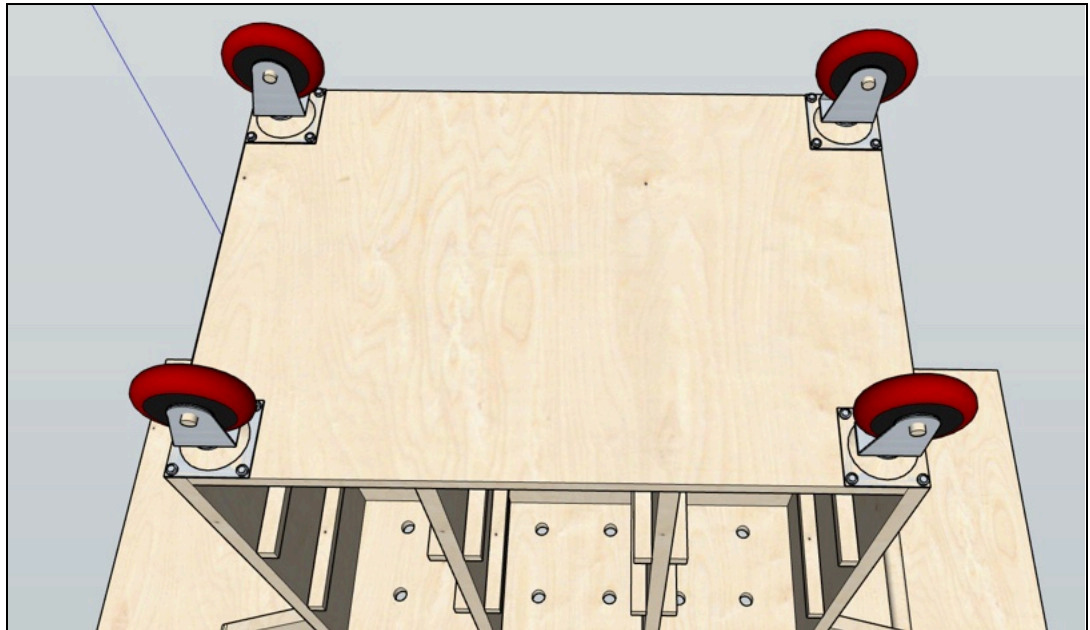


- k. Flip the cart back right-side-up.
- l. Install the Eaves to the Top with the sewing machine table hinges. To do this is a bit challenging. Find instructions for installing the hinge, which will require tracing out the desired location for the hinges and then routing (with a router or rotary tool) or chiseling out the profile needed to set the hinges into the wood. Install each of the four hinges 4" from the sides.



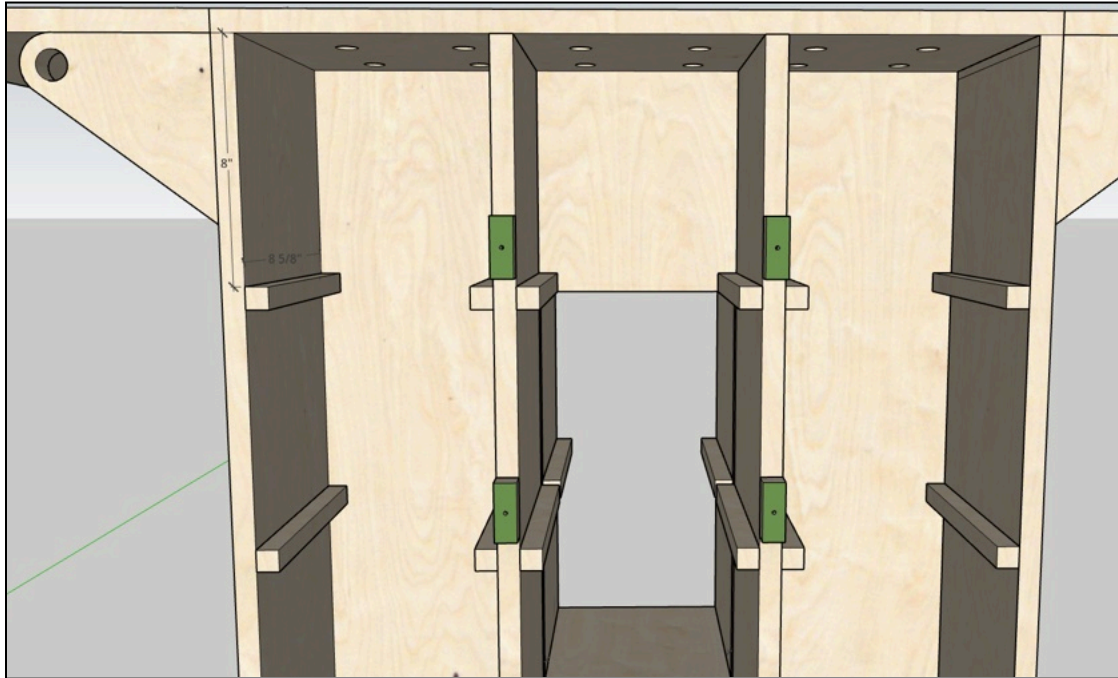


- m. Flip the cart back upside down. So that the bottom is facing up.
- n. Install the casters following the manufacturers directions. If needing to pre-drill or drill holes through the Bottom for nuts and bolts, do so at this time and firmly affix the casters. Countersink as necessary to avoid bolts from impeding the drawers/boxes. Note: if only 2 of the casters swivel, it is recommended to install them on the front (non-handle) side of the cart to make steering easiest.



- o. Sand the cart, especially the corners and edges using the fine sandpaper (400 grit or similar) until there are no sharp edges to sliver.
 - p. Follow the manufacturers directions to apply the wipe-on polyurethane over all wooden surfaces. 2 coats minimum is recommended.
7. Use the adhesive squares or circles to affix to the hinged Eave Supports where they contact the End of the cart when folded in. This will keep the Eave Supports from flapping while the cart is in motion.
 8. At this point, select what kind of boxes, draws or shelves you want on the cart and construct them. Instructions for a standard box are provided below in this plan.

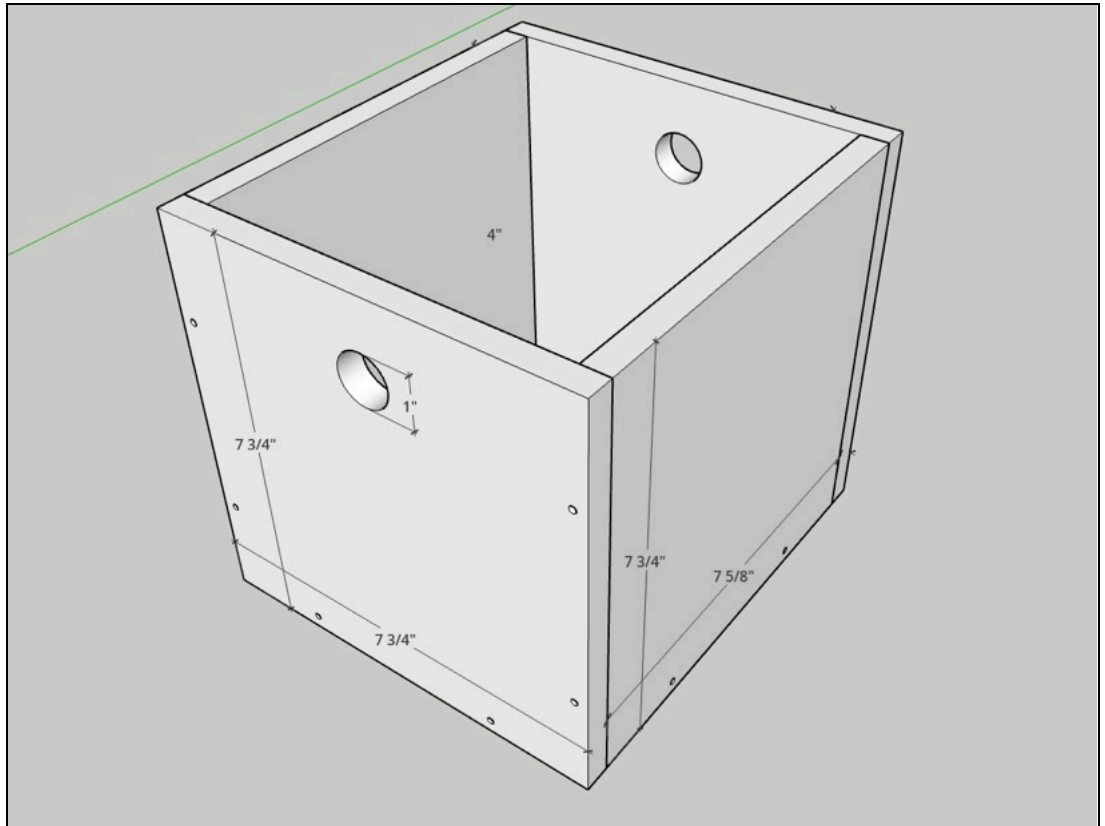
9. One note on retaining the boxes. Without drawer stops to secure the boxes, they will wiggle loose while the cart is rolled. To prevent that, it is recommended to cut out wooden pieces $\frac{3}{4}$ " by $\frac{3}{4}$ " by 2" from the remaining pieces of the plywood sheet and then attach them to the Vertical Supports between the boxes with a #8 $1\frac{1}{2}$ " screw snug (tight enough to hold them in selected position but loose enough that they can be turned). See the diagram (Drawer Stops in green). A total of 12 are needed if all slots are filled with Drawer Boxes. If filling the slots with other shelving or containers of your choice, adapt as needed. Bungees or other materials can be used to secure items in the slots.



Building the Drawer Boxes

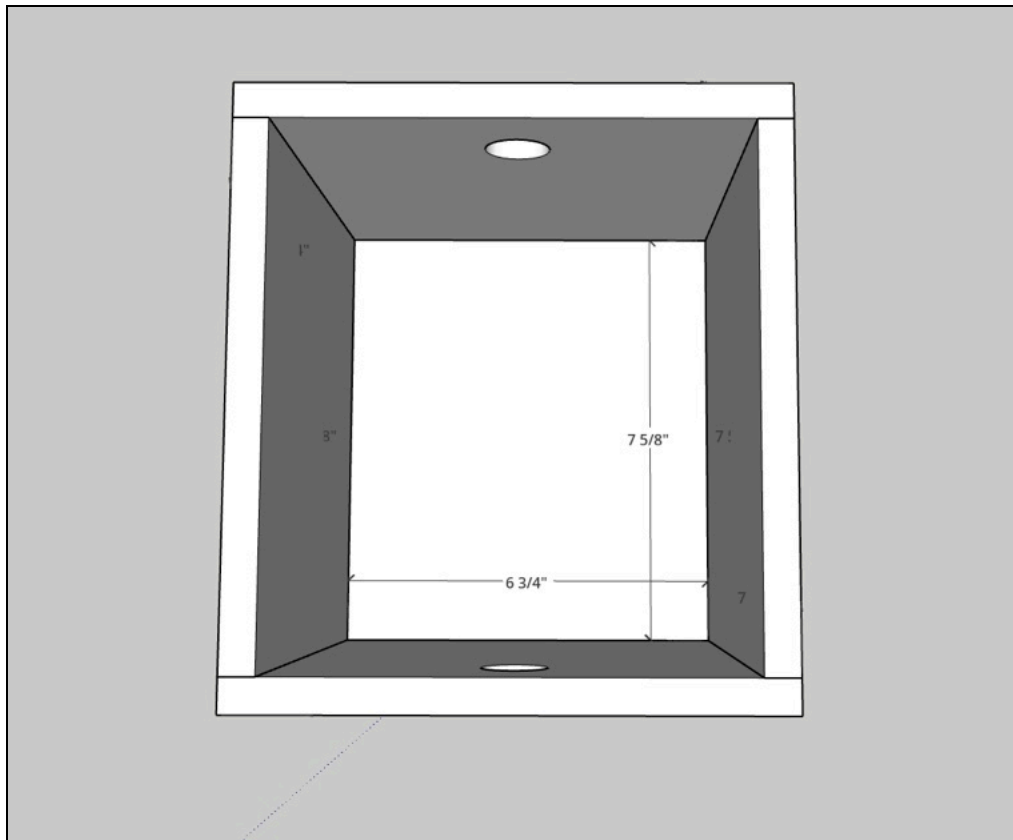
With a single sheet of $\frac{1}{2}$ " thick 4' x 8' birch plywood, you can create a set of boxes to function as drawers for the Fix It Cart and customize as you need.

Cut out according to the dimensions shown here. It is recommended to layout your sheet such that you can cut multiple of the sides and bases in the fewest number of passes.



Cutting

1. The fronts and backs of the boxes are $7\frac{3}{4}$ " x $7\frac{3}{4}$ " square to make cutting easy. 2 per box.
2. The sides are $7\frac{3}{4}$ " x $7\frac{5}{8}$ ". 2 per box.
3. The bottom is $6\frac{3}{4}$ " x $7\frac{5}{8}$ ". 1 per box.



Assembly

1. Using a 1/16" drill bit, drill holes in roughly the spots indicated on the drawing above on the front and back and the two holes near the base of the side pieces. The holes should be 1/4" from the edge. No need to pre-drill into the actual sizes. You can do this ahead of time to make final assembly easier.
2. On one or both of the front and back pieces, drill a 1" hole using the forstner bit. This will function as the handle for the box. It should be drilled at least 1" down from the top edge and on the centerline as shown. Customize as desired.
3. Screw the box together using the #6 screws, following the pattern dictated by the pre-drilled holes. See the diagram above for clarity.
4. Sand and finish with the polyurethane the same way as for the cart.

Supplement: Smaller Cart Alternative

Making the smaller cart is the same as the larger cart except for the height of the cart, the number of drawers to build and the interior center panel.

Build Instructions

To make the smaller cart, follow all the instructions for the cart above with the following exceptions and different dimensions. Each smaller cart will also require a sheet of $\frac{3}{4}$ " x 4' x 8' birch plywood.

1. The height of the End pieces is $17 \frac{1}{2}$ " (width remains 18")
2. The height of the Vertical Support pieces is $16 \frac{3}{4}$ ". Width remains the same.
3. In lieu of the Inner Wall, cut two pieces measuring $16 \frac{3}{4}$ " x $8 \frac{1}{2}$ ". They will be positioned the same as the Inner Wall of the larger cart, but such that there is a gap of one slot space between them.
4. Make fewer shelf supports since you only need 12 per cart for the smaller variant.

Appendix A: Example Materials

Brass Sewing Machine table hinge / Butler Tray hinge:



Can generally be purchased online or in specialty woodworking shops.

3" x 2" Brass Utility Hinge



Available in most hardware stores or online

5" Lockable Casters, 2 swiveled and 2 fixed



Generally, only available online. This example was purchased from Amazon:
https://www.amazon.com/dp/B0BR8848D2?ref_=ppx_hzod_title_dt_b_fed_asin_title_0_0